ML student capstone project

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Introduction

I chose for this project the IMDB dataset, so I can make my model understand and detect human opinions

Dataset introduction

My code found a few key insights about the dataset, and they are that : the dataset has 50000 rows, two columns (positive & negative) and that the balance between the two columns is balanced (positive = 25000 reviews & negative = 25000 reviews) and when trying to clean the dataset I saw that the dataset didn't need any cleaning since the there wasn't any missing data in the set we didn't need to do any cleaning

Models used

I trained three models (other than the neural network, more about that later in the report) and they are : 1 - Logistic Regression, 2 - KNN (K's nearest neighbors) and 3 - Random forest

I put those 3 models through 4 different metrics (Accuracy, Precision, Recall and F1 score) and the scores for the models through the metrics is in the table below (Shape 1)

X.XXXXXX format | Shape 1	Logistic Regression	Random Forest	KNN
Accuracy	0.889200	0.822700	0.742100
Precision	0.882769	0.789833	0.717520
Recall	0.897600	0.879400	0.798600
F1 Score	0.890123	0.832213	0.755892

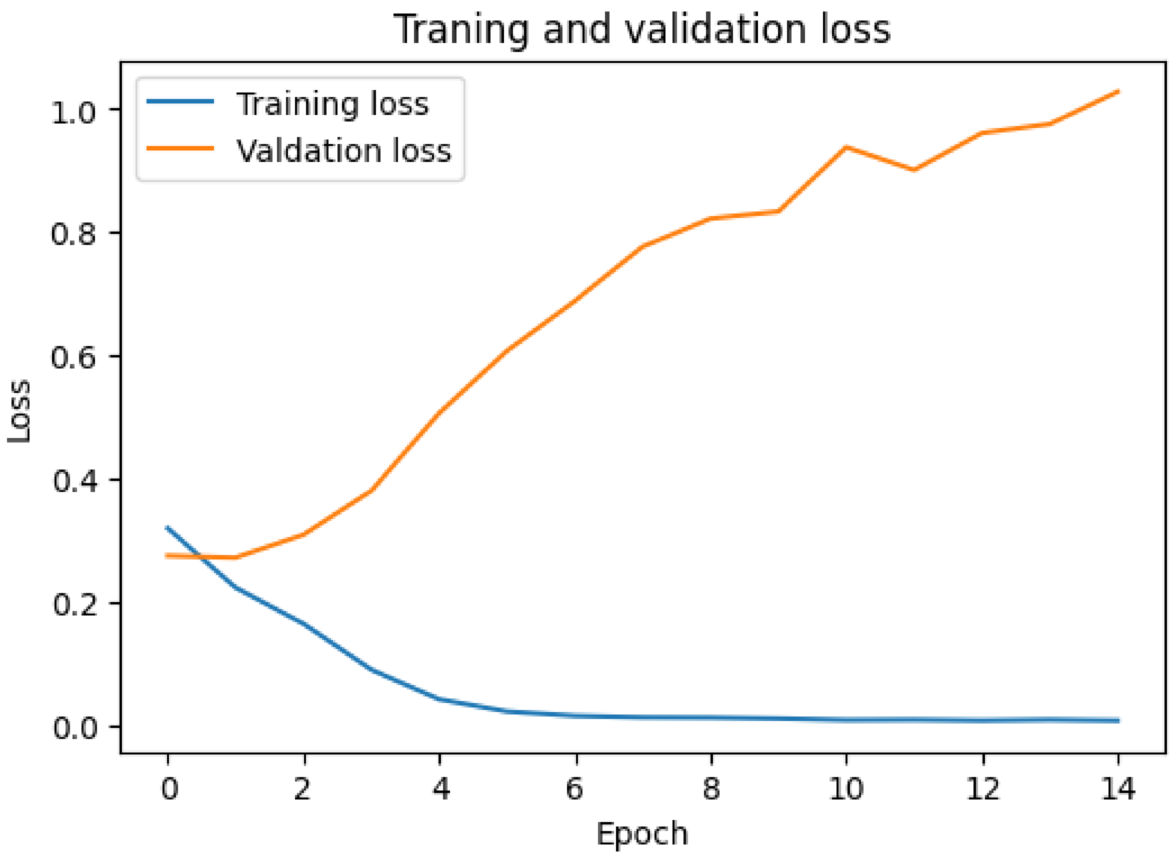
Neural Network

My neural network had to dense layers (128 and 64 neurons) with dropout. I tested training with multiple number of epochs and landed on a number of epochs that is not too good but has good stats, and it is 15 epochs. Here are the metrics results for it (Shape 2)

X.XXXXXX format | Shape 2	Neural Network
Accuracy	0.870700
Precision	0.890786
Recall	0.845000
F1 Score	0.867289

Here is the Training and Validation test loss for the Neural Network (Shape 3) :

Shape 3



Results & Comparison

Comparing these results with the other models we find that the Logistic regression is the best model in every metric but the precision metric, that goes to our neural network and that KNN is the worst model in all metrics

Here is the difference between every model :

Accuracy :

1 - Logistic Regression. --(0.018500)-- 2 - Neural Network. --(0.048000)-- 3 - Random Forest. --(0.080600)-- 4 - KNN.

Precision :

1 - Neural Network. --(0.008017)-- 2 - Logistic Regression. --(0.092936)-- 3 - Random Forest. --(0.072313)-- 4 - KNN.

Recall :

1 - Logistic Regression. --(0.018200)-- 2 - Random Forest. --(0.034400)-- 3 - Neural Network. --(0.046400)-- 4 - KNN.

F1 Score :

1 - Logistic Regression. --(0.022834)-- 2 - Neural Network. --(0.035076)-- 3 - Random Forest. --(0.076321)-- 4 - KNN.

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again here are the results :

X.XXXXXX format	Logistic Regression	Random Forest	KNN	Neural Network
Accuracy	0.889200	0.822700	0.742100	0.870700
Precision	0.882769	0.789833	0.717520	0.890786
Recall	0.897600	0.879400	0.798600	0.845000
F1 Score	0.890123	0.832213	0.755892	0.867289

What I learned

More Complex models aren’t alway better than simpler models, cleaning text before traning makes the biggest difference to accuracy and that if you do basically anything with the data before making the training, Validation and test (or training, test split) the data will be basically ruined and you need to start once again

Eyadalmiman/**Student-Capstone-Project-ML-**

This is the student capstone project Machine learning edition started on May 22nd Friday

1 Contributor   0 Issues   1 Star   0 Forks